Cross-species ncRNA annotation using synteny-constrained embedding similarity

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Abstract

Non-coding RNAs (ncRNAs) are poorly conserved at the sequence level, which makes cross-species annotation by homology unreliable. We present CURIA (Cross-species Unified ncRNA Inference and Annotation), a pipeline that projects reference ncRNA loci into a query genome along whole-genome alignment chains and resolves annotation in the embedding space of an RNA language model rather than by raw sequence identity. CURIA refines short ncRNAs by local matching in embedding space and analyses long ncRNAs through localized structured subregions ("islands"), restricting the genome-wide search to syntenic candidate loci. The current implementation uses RiNALMo embeddings, whose near-invariance to flanking context enables each island to be embedded once and aligned at nucleotide resolution. We evaluate CURIA on several mammalian genome pairs. *[Quantitative results to be inserted.]*

1 Introduction

Non-coding RNAs carry out diverse regulatory and structural roles, yet they evolve under weak sequence constraint, so orthologous ncRNAs are frequently undetectable by sequence-alignment search. Structure and syntenic position are often better preserved than primary sequence. CURIA exploits both: it uses whole-genome alignment chains to constrain *where* an orthologous ncRNA can be, and an RNA language model to decide *whether* a candidate locus carries a conserved signal.

This work builds on orthology inference at genomic scale (Kirilenko et al., 2023) and on RNA foundation models (Penić et al., 2024; Chen et al., 2022). Earlier prototypes of CURIA used RNA-FM; the present version uses RiNALMo, motivated by its context stability (Section 2).

Contributions.

- A synteny-constrained pipeline that projects reference ncRNA loci into a query genome and resolves them in embedding space.
- An island-based representation for long ncRNAs that localizes conserved structured subregions.
- An embed-once, align-at-nucleotide-resolution matching procedure enabled by RiNALMo's near-invariance to flanking context.

2 Methods

2.1 Overview

CURIA takes a reference annotation (BED12), reference metadata, a query and a reference genome (2bit), and whole-genome alignment chains, and produces ncRNA annotations in the query genome. Processing has three stages: (i) orthology-guided locus projection along chains, (ii) embedding of candidate loci with an RNA language model, and (iii) resolution of the annotation — short ncRNA refinement by local matching, or long ncRNA analysis via islands.

2.2 Synteny-constrained locus projection

Reference loci are projected into the query genome using alignment chains, restricting the search to syntenic candidate regions (Kirilenko et al., 2023).

2.3 RNA language-model embeddings

Candidate loci are embedded with RiNALMo (`giga-v1`; 1280-dimensional token embeddings) (Penić et al., 2024). Unlike RNA-FM (Chen et al., 2022), RiNALMo embeddings are essentially invariant to flanking sequence: moving the input boundaries leaves token embeddings almost unchanged (negligible MMD drift). This context stability is what allows the matching stage below to embed each region once.

Two linear projections are fit on reference embeddings and stored as pipeline artifacts: a k=16 PCA for position-level matching and a k=64 PCA for island finding. A logistic-regression classifier separates conserved ("signal") from background ("noise") positions during island finding.

2.4 Short ncRNA refinement

Short ncRNAs ($\leq$ 160 bp) are refined by local matching of query and reference embeddings, reporting the best-scoring alignment per locus.

2.5 Long ncRNA islands

Long ncRNAs are represented by localized structured subregions ("islands"). Because embeddings are context-stable, each island is embedded once and aligned to the reference by Smith–Waterman on the per-token cosine similarity dotplot, yielding nucleotide-resolution matches without per-window re-embedding.

2.6 Implementation and parameters

Per-model parameters (PCA files, signal/noise classifier, scan and match thresholds, embedding strategy) are selected automatically by the `--model` flag and are recorded in the model registry. Embeddings use PyTorch's memory-efficient scaled-dot-product attention; runs were executed on GPU (CUDA / MPS).

2.7 Evaluation

CURIA was evaluated on several mammalian genome pairs, including human–mouse.

3 Results

3.1 Context stability of RiNALMo vs RNA-FM

3.2 Short ncRNA recovery

3.3 Long ncRNA islands and conservation

4 Discussion

CURIA resolves ncRNA orthology in embedding space rather than by raw sequence identity, constrained to syntenic loci. Moving from RNA-FM to RiNALMo removes the per-window re-embedding that RNA-FM's context sensitivity forced, making island alignment both faster and more accurate on true matches.

Limitations.

Availability. CURIA is available at `https://github.com/kirilenkobm/CURIA_pipeline`.

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Author contributions

References

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